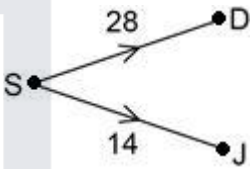
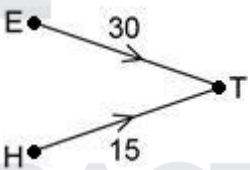


Mark schemes

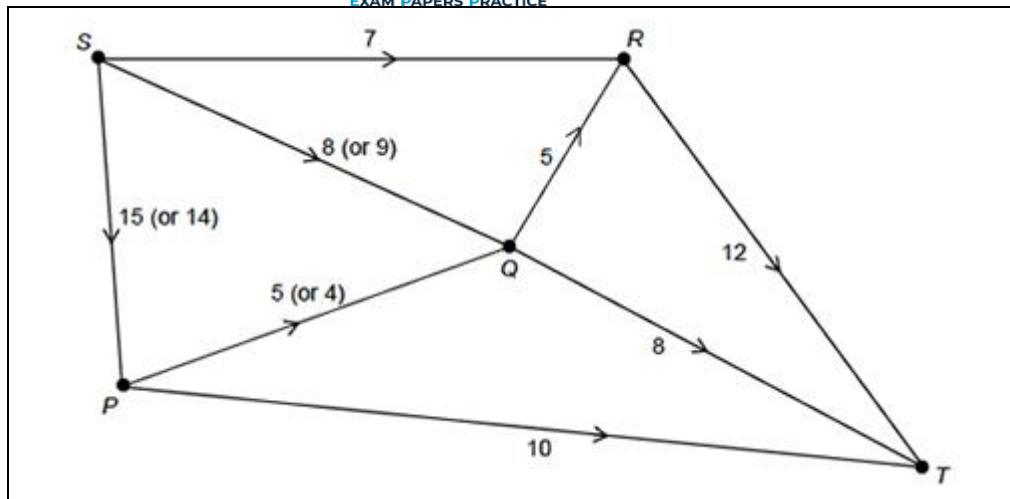
Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds the value of the cut	AO1.1b	B1	37
(i)				
(ii)	Deduces a weak inequality for the value of max flow based on their value of the cut Must see $\leq$ their value of the cut OE	AO2.2a	B1F	Max flow $\leq$ 37
(b)	Lists both sources CAO	AO1.1b	B1	<i>D</i> and <i>J</i>
(i)				
(ii)	Adds supersource <i>S</i> with directed arcs to <i>D</i> and <i>J</i> labelled with weights at least 28 and at least 14 respectively	AO1.1b	B1	
(c)	Lists both sinks CAO	AO1.1b	B1	<i>E</i> and <i>H</i>
(i)				
(ii)	Adds supersink <i>T</i> with directed arcs to <i>E</i> and <i>H</i> labelled with weights at least 30 and at least 15 respectively	AO1.1b	B1	
				Total 6 marks

©2025 Exam Papers Practice. All rights reserved.

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Determines correctly the value of both cuts	AO1.1b	B1	35 and 36
(b)	Deduces the maximum flow in 'their' network by use of the maximum flow-minimum cut theorem and explains their reasoning.	AO2.4	E1	Max flow = 30 as max. flow = min. cut and the value of the minimum cut is 30



(c)	Determines correctly the flow along the saturated arcs	AO1.1b	B1	$SR = 7, QR = 5, RT = 12, QT = 8, PT = 10$
	Determines correctly a possible flow for each of SP, SQ and PQ	AO1.1b	B1	$SP = 15$ and $SQ = 8$ and $PQ = 5$ or $SP = 14$ and $SQ = 9$ and $PQ = 4$
Total 4 marks				

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)				
(i)	Explains correctly using maximum flow into and minimum flow out of A	AO2.4	E1	Max flow into vertex $A = 7$ Min flow out of vertex $A = 4 + 3 = 7$ As flow into $A =$ flow out of A , both AD and AB must be at the lower capacity
(ii)	Explains correctly using maximum flow into and minimum flow out of E	AO2.4	E1	$AD = 4$ from (a)(i), so $DE = 1$ and $DT = 3$. Since flow out of vertex E is at least 2, and $DE = 1$, BE must be at its upper capacity of 1.
(b)	Explains correctly the statement using a cut in the network	AO2.4	E1	Min flow across the cut $\{S, A, B, C\} / \{D, E, G, T\}$ $= AD + BE + \min(BG) + \min(CG)$ $= 4 + 1 + 5 + 2$

			$= 12 > 11$
--	--	--	-------------

(c)
(i)

Finds a potential increase and decrease for each arc with a value on the two arrows for each arc, with SA, SC, AB, BC and CF correct.

AO1.1a

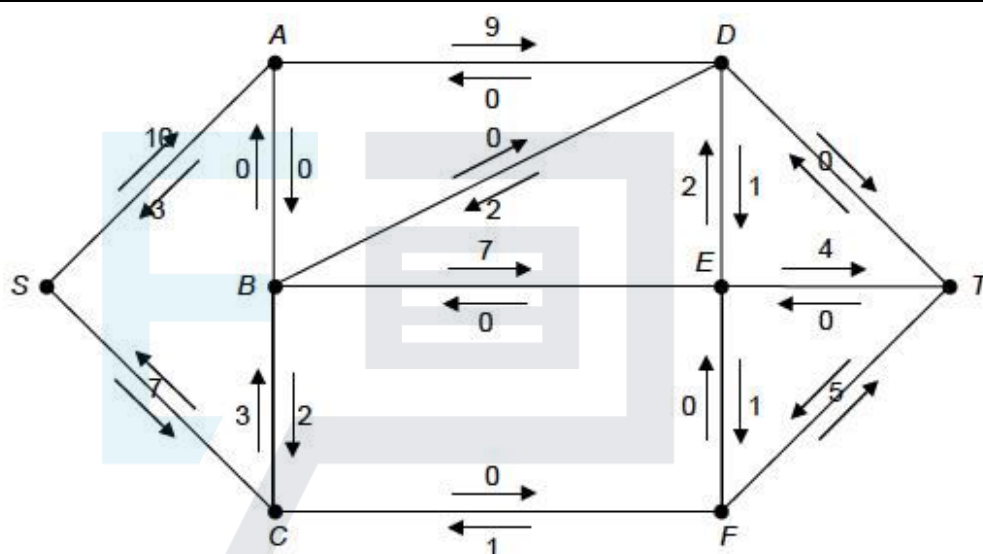
M1

Determines correctly the potential increase and decrease for each arc with values on all arrows correct.

AO1.1b

A1

See diagram below



(ii)

Finds correctly one augmenting path and flow

AO1.1b

B1

Finds correctly two augmenting paths and the flows

AO1.1b

B1

Augmenting Path	Flow
SADEFT	1
SCBET	3
SADBET	1

Finds correctly three (or four) augmenting paths and the flows, and clearly states the maximum flow in context with units

AO3.2a

B1

Maximum flow through the network is 22 litres per second

(iii)

Constructs a rigorous mathematical proof using the value of a cut and the maximum flow-minimum cut theorem (to achieve this mark, the student must clearly show the cut they are calculating the value of **and** clearly state that the value of this cut is equal to the value of the

AO2.1

R1

$\{S, A, B, C, D, E\} / \{F, T\} =$
 $7 + 3 + 9 + 3 = 22$
 As the flow (22) is equal to the value of the cut (22), the maximum flow is 22 by the maximum flow-minimum cut theorem.

flow in (c)(ii) and then conclude that the flow is maximal by the maximum flow-minimum cut theorem)			
--	--	--	--

(iv)	Argues that, as 7 litres per second are initially flowing through node <i>E</i> , only a further 2 litres per second can flow through node <i>E</i>	AO2.4	R1	Flow through network can only increase by 2 litres per second as <i>DT</i> and <i>CF</i> are already saturated Therefore restricted capacity node reduces the maximum flow to $17 + 2 = 19$ litres per second
	Interprets the impact of the restricted capacity node, concluding that the maximum flow is reduced to 19 litres per second	AO3.A	R1	
				Total 11 marks

Q4.

- (a) (i) Max Flow = 50
(Min cut = 50)
Either statement

E1

- (ii) $35 \leq \text{max flow} \leq 50$
(or min cut)
E1 for strict inequalities

E1, E1

- (iii) Error or contradiction
oe

E1

4

- (b) At *F*,

$$\left. \begin{array}{l} \text{flow in} \geq 8 \\ \text{flow out} \leq 7 \end{array} \right\}$$

*Stating *F* and one of the 'flows'*

M1

A1

2

Q5.

(a) $ABEH$ 8

$ACFH$ 5

$ADGH$ 11

B1

1

(b) (i) $ACEH$ 2

One correct route and flow

M1

$ACGH$ 4

At least one other correct

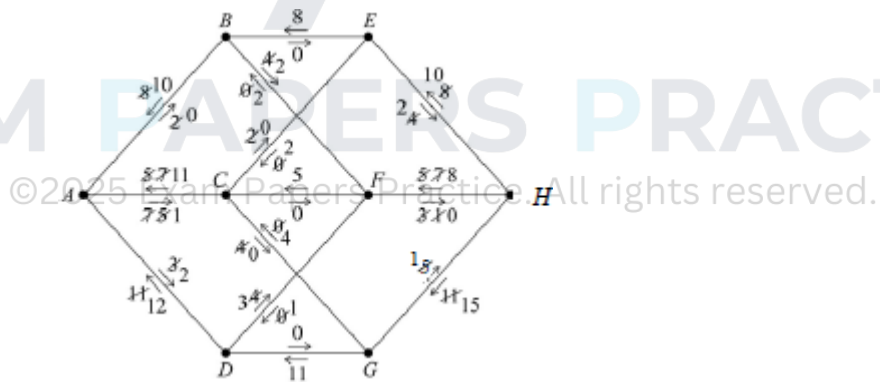
A1

Either $ADFH$ 1 and $ABFH$ 2

All correct

A1

Or $ADFH$ 3



Forward and back flows on diagram

M1

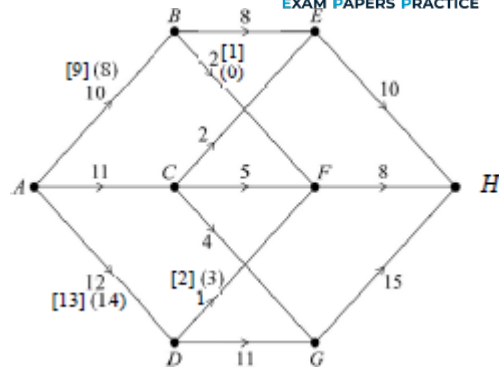
All correct

A1

5

(ii) Max flow 33

B1



OE

B1

2

(c) Cut through BE, CE, FH, CG, DG

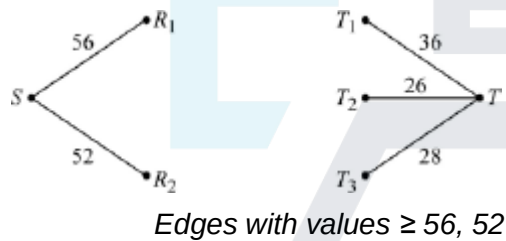
B1

1

[9]

Q6.

(a)



Edges with values $\geq 36, 26, 28$

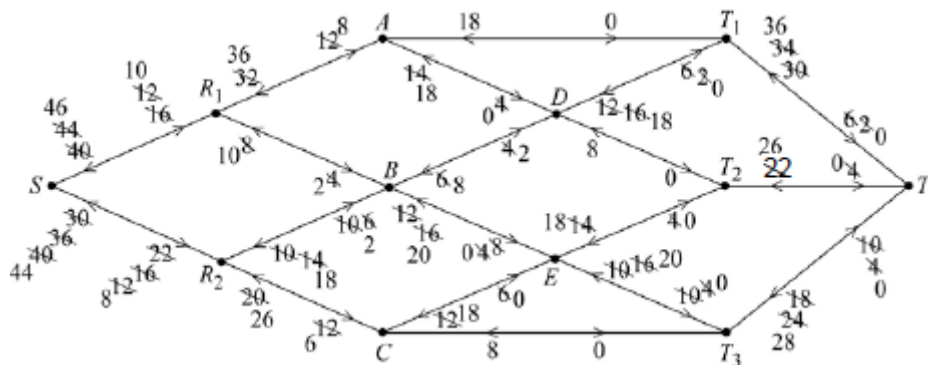
B1

©2025 Exam Papers Practice. All rights reserved.

B1

2

(b)(i)



initial diagram with forward / back flows

M1

Fully correct diagram

A1

$S R_1 A D T_1 T$ 4

One correct path and flow

M1

$S R_1 B D T_1 T$ 2

At least one other correct path and flow

A1

$S R_2 C E T_3 T$ 6

$S R_2 B E T_2 T$ 4

all correct

A1

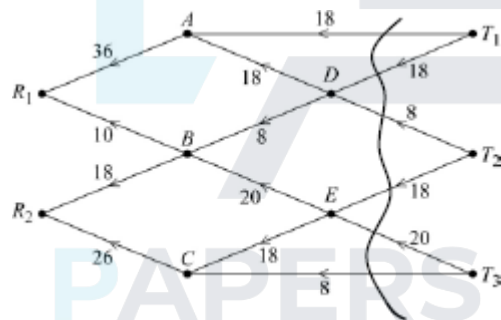
$S R_2 B E T_3 T$ 4

(ignore connections to S and T)

5

(ii) Max flow 90

B1



©2025 Exam Papers Practice. All rights reserved.

B1

2

(c) Cut through (shown)

PI by correct list

B1

$AT_1, DT_1, DT_2, ET_2, ET_3, CT_3$

B1

OE

2

[11]

Q7.

(a) $10 + 13 - 1 + 17$

3 values added and -1 (condone one slip)

M1

= 39

A1

2

(b) (i) $DE = 12$
on Figure 1

B1

$FG = 7$

B1

2

(ii)

arc	forward	backward
SA	3	1
AB	1	1
BT	0	1
SC	0	2
CA	0	1
AD	0	1
CD	1	1
DE	1	2
BE	1	3
ET	2	3
SF	1	1
FC	1	2
FD	1	0
FG	1	1
DG	0	1
EG	2	1
GT	1	3

©2025 Exam Papers Practice. All rights reserved.

at least 6 pairs correct on Figure 2 (must have arrows)

M1

all correct

A1

2

(iii) **Table**

Path	Extra Flow
$SABET$	1
$SFDGT$	1
$SACDGT$	1

1 correct path and extra flow

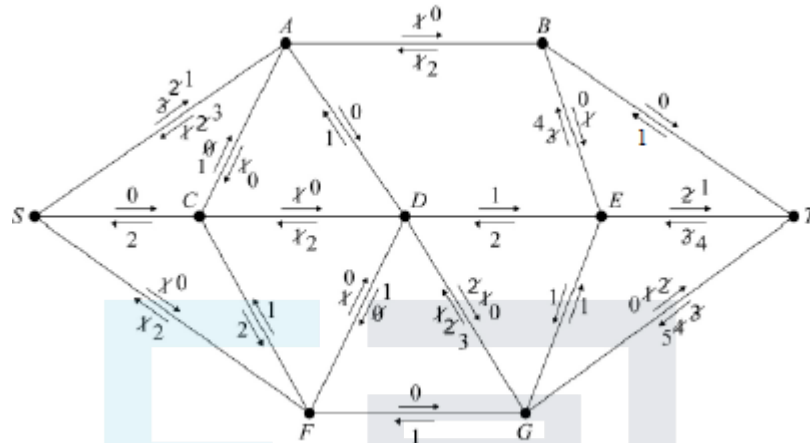
M1

all correct

DEG triangle may have different flows with implications to triangle GET.

A1

Network



1 path correctly augmented forward and backward
but must have earned M1 in part (b)(ii)

M1

network correct

A1

4

(c) (i) Max flow = 37

B1

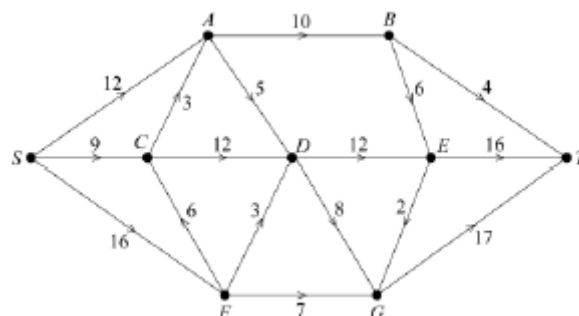
©2025 Exam Papers Practice. All rights reserved.

1

(ii) Max flow

correct flow of 37

B2



condone 2 slips or omissions in flow of 37 **or** "correct"
feasible flow of 36
for SC1

2

- (d) Cut through AB , AD , CD , FD and FG
 $\{S, A, C, F\} \{B, D, E, G, T\}$

B1

1

[14]

Q8.

- (a) (i) $18 + (0+) 10 + 3 + 5 \quad (= 36)$

B1

1

- (ii) 30, 32, 36 (missing cut values)
B1 each value correct

B3

3

- (iii) Max flow = 29
Award B0 E1 if their $\min^m$ cut is < 29

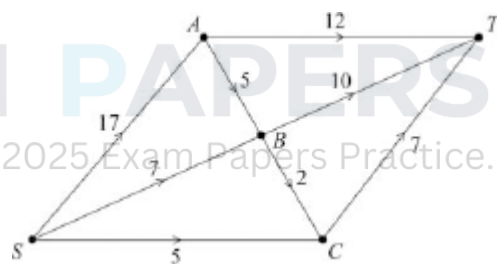
B1

because value of minimum cut is 29
and $\min^m$ value explained as max flow

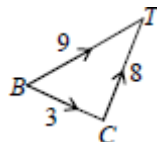
E1

2

- (iv)



may have



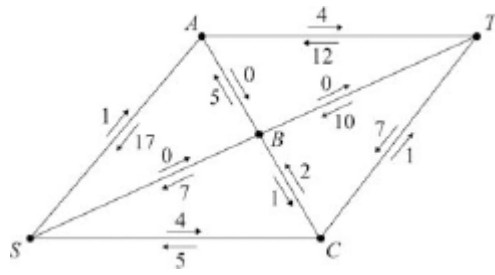
*see alternative
solution on next
page*

B1

cao

2

(b) (i)



potential flow (forward and back)
4 pairs 'correct' including SC and AT
ft their (a)(iv) provided $0 < \text{flow} < 30$

M1

all pairs correct (condone missing 0s)
ft their (a)(iv) if correct flow < 29

A1 ✓

one correct flow in table

m1

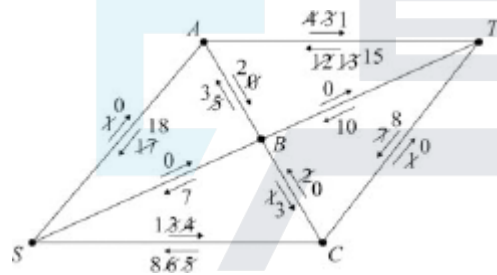


table
correct

SAT	1
SCT	1
SCBAT	2

©2025 Exam Papers Practice. All rights reserved.

If (a)(iv) flow < 29 then may score A1 for correct table
giving max flow of 33

A1

(see also the alternative solution)
modifying flows (forward and back)
1 flow correct ft their initial flow

m1

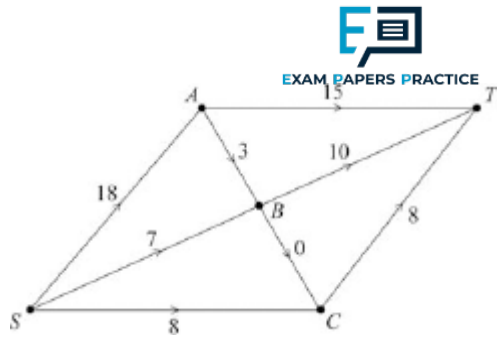
modified flows all correct, including all 0s (may score A1
from a correct flow < 29 seen in (a)(iv) if final flow correct)

A1

6

(ii) new max flow = 33

B1



6 flows correctly interpreted from their labelling procedure
provided M2 or M3 scored in (b)(i)
(may have AB 2, AT 16, BT 9 – see over)

M1

flow correct

SC B1 if flow of 33 shown correctly but not from correct
labelling procedure

A1

3

[16]

Q9.

- (a) $SP \geq 12$
 $SQ \geq 10$
 $SR \geq 17$

S in correct place, (arrows) and capacities

B1

- $YT \geq 18$
 $ZT \geq 17$

T in correct place, (arrows) and capacities

B1

2

- (b) $SPUYT$ 10

B1

- $SRVWZT$ 8

B1

2

- (c) (i) Initial flow forward and backward

PU 2 and 10 ; UY 0 and 10

B1

RV 0 and 8 ; VW 1 and 8 ; WZ 2 and 8



B1

2

- (ii) Two correct routes and flows on Figure 2

<i>SPUYT</i>	10
<i>SRYWZT</i>	8
<i>SPUXYT</i>	2
<i>SQVUXYT</i>	6
<i>SRWXZT</i>	5
<i>SRWZT</i>	2

(other possibilities)

M1

Correct additional flows

Max flow = 33

A1

Adjustment of at least 4 edges corresponding to flows
(forward and backward)

M1

Correct final flows forward and backward
(must score A1 for table)*edges UY, UX, WX and WZ will be saturated**XY + XZ = 13 in back flow*

A1cso

4

- (d) Cut with value 33 is through
-
- UY, UX, WX and WZ*

B1

1

[11]

Q10.

- (a) (i) Value of cut =
- $38 + 25 + 0 + 0 + 34$
-
- Must show correct addition*

= 97

AG

B1

1

- (ii)
- $\{S, A\}, \{B, C, T\}$
- 65

B1

$\{S, B\}, \{A, C, T\}$ 57

B1

$\{S, B, C\}, \{A, T\}$ 72

B1

$\{S, A, B, C\}, \{T\}$ 56

B1

4

- (iii) Maximum flow = 53
ft their least cut value

B1ft

Minimum cut = Max flow

E1

2

- (iv) Their max flow on SA, SB or AT, CT

M1

All correct

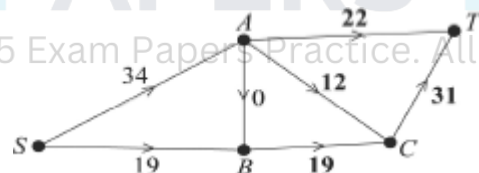
A1

2

AT 22; AC 12; BC 19; CT 31

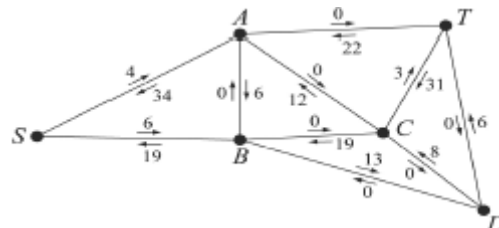
$AB = x$; $AS = x + 34$; $SB = 19 - x$

$0 \leq x \leq 4$



©2025 Exam Papers Practice. All rights reserved.

- (b) (i) Initial flow on Figure 3
Forward potential and backward flow
Condone 2 slips, ft their Figure 2



M1

One correct augmented path in table and correct flow

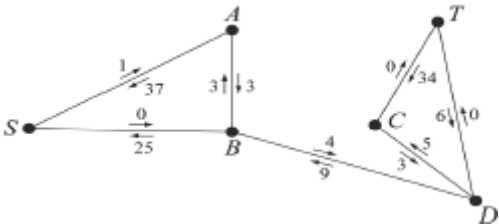
Path	Additional Flow
<i>SBDT</i>	6
<i>SABDCT</i>	3

M1

Table correct with total additional flow = 9

A1

Final network correct with evidence of labelling procedure used



A1

4

(ii) New maximum flow = 62

B1

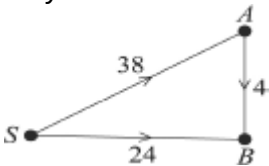
Correct maximum flow on network



B1

©2025 Exam Papers Practice. All rights reserved.

May have



2

[15]

Q11.

- (a) Value of cut = $10 + 10 + 15 - 4 - 1$
condone one slip if working shown

M1

= 30

A1

2

- (b) *BT 2, DE 3, ET 12*
any 2 correct

B1

all correct

B1

2

- (c) (i) Initial flows forward and back or double
 Arc with arrows (at **least 6 pairs correct**)
Condone pairs of values, (coordinates)
with single arrow

M1

*all correct (condone pairs with single
 arrow provided key indicated)*

A1

2

(ii)

Path	Flow
<i>SABT</i>	2
<i>SCDET</i>	1
<i>SACBT</i>	1

(or SCBT instead of SACBT with flow 1
first correct path and flow

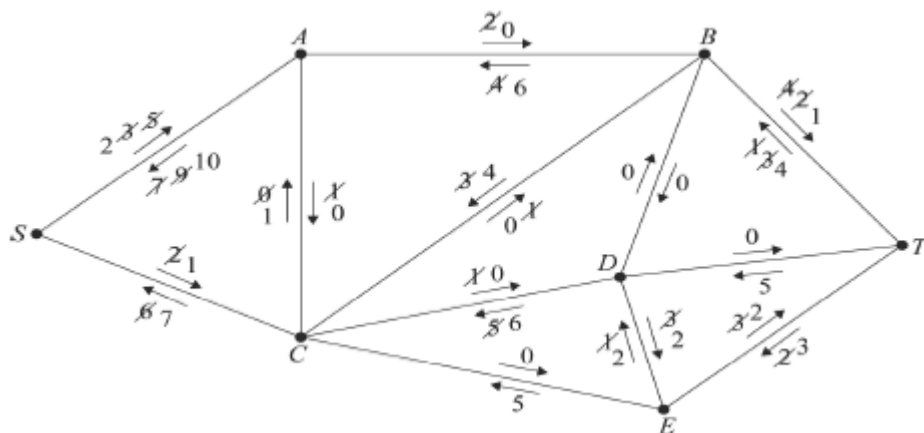
M1

another correct path and flow

A1

all correct
(other possibilities also)

A1



Must have forward and backward flows
augmenting flows (6 pairs correct)

M1

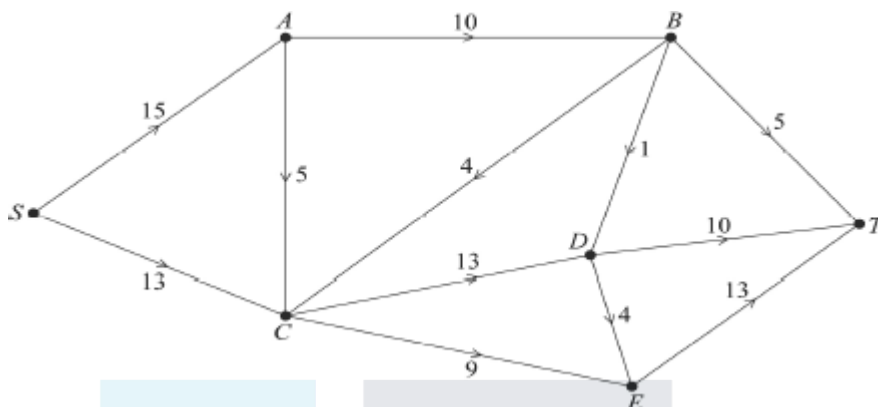
correct

A1

Alternative SA (3 & 9) SC (0&8)

5

(iii)



May have
SA (14), SC (14)
and AC (4) using
alternative
Maximum flow values

at least 8 correctly interpreted from their
Figure 3 but $24 < \text{their maxflow} < 29$

M1A1

2

(d) Cut through AB, CB, CD and CE

May use $\{S, A, C\} / \{B, D, E, T\}$

But must have total flow of 28 in their
network (condone one slip)

©2025 Exam Papers Practice. All rights reserved.

B1B1

1

[14]

Q12.

(a) Value of cut = $30 - 10 + 12 + 20$

M1

= 52

Full marks for correct answers without working

A1

2

(b) $AE = 9$;

B1

$$EF = 5;$$

B1

$$FG = 4$$

B1

3

(c) (i) Attempt at forward and backward flows

SA 2 & 4; AB 1 & 3; BT 1 & 3

SD 3 & 1; DA 0 & 3; AE 0 & 3

BE 0 & 7; DE 2 & 0; ET 1 & 3

FD 2 & 1; EF 5 & 1; EG 1 & 5

FG 1 & 2; GT 3 & 0

At least 5 pairs correct

M1

10 pairs correct

A1

all correct

A1

3

(ii) First flow augmenting path and correct flow on table

M1

Table correct

A1

Adjusting flows – forward and back

©2025 Exam Papers Practice. All rights reserved.

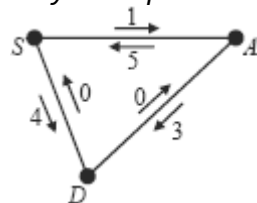
M1

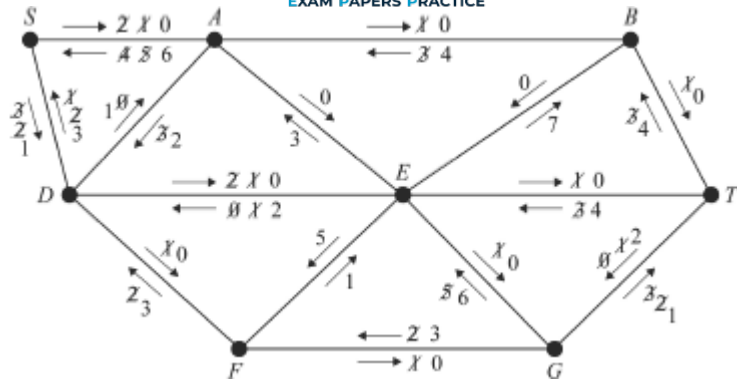
Correct

A1

4

May end up with





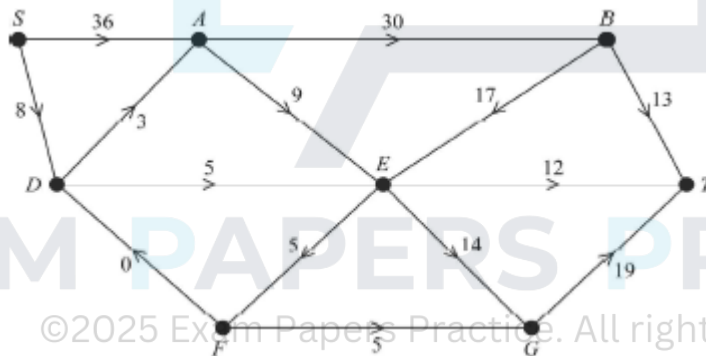
Path	Extra flow	
SABT	1	Or SDET
SADET	1	
SDFGT	1	
SDEGT	1	

- (d) Max flow of 44 shown on figure 3
up to 2 slips

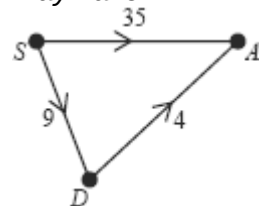
all correct

M1

A1



May have



2

- (e) Cut through their saturated arcs

M1

Cut passes through AB, AE, DE and DF
Or BT, ET, EG, and FG

A1

2

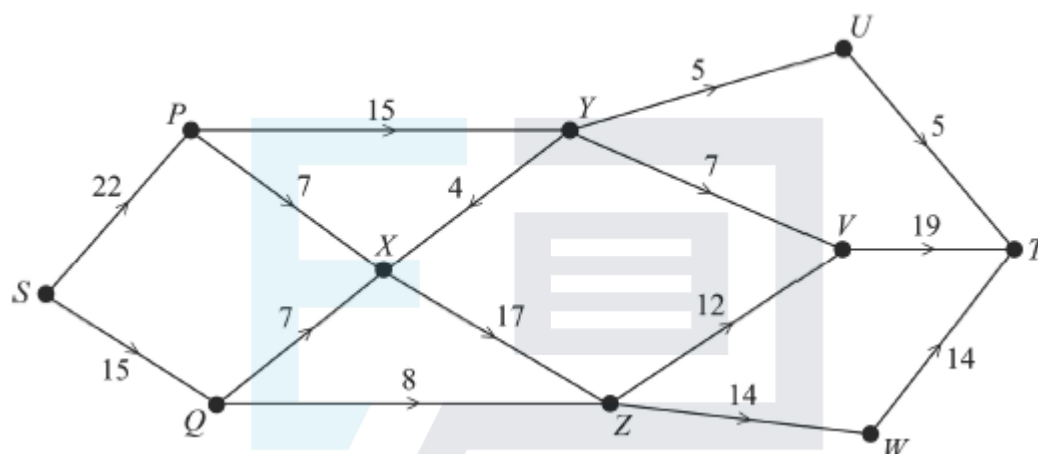
Q13.

- (a) Correct position of S and T

M1

Values on edges SP , SQ , UT , VT and WT

A1



2

- (b) (i) Cut C has value 40
 $15 + 0 + 17 + 8$

B1

1

©2025 Exam Papers Practice. All rights reserved.

- (ii) Max flow ≤ 40

E1

1

- (c)

Route	Flow
$SQZWT$	8
$SPYXZVT$	4

B1B1

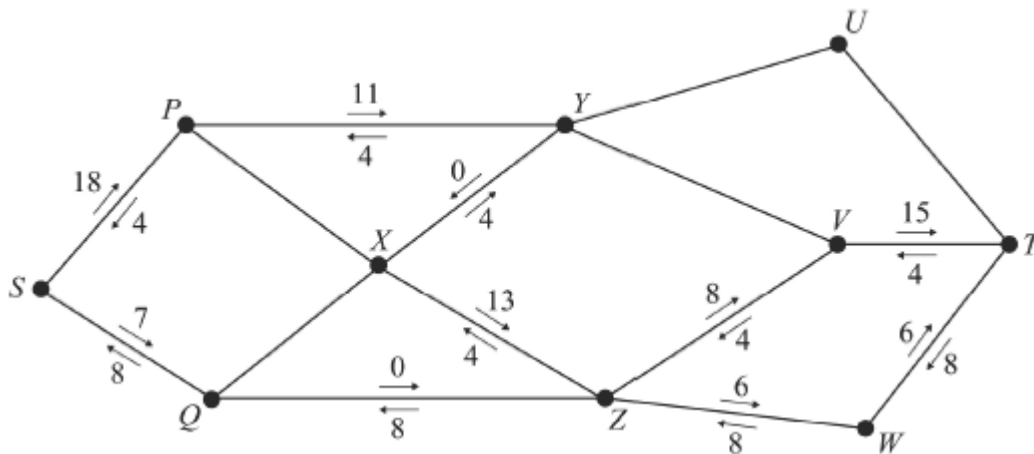
2

- (d) (i) 3 forward and backward flows correct

M1

All initial values correct on edges below

A1



2

(ii)

Route	Flow
SQZWT	8
SPYXZVT	4
SPYUT	5
SPYVT	6
SPXZVT	7
SQXZWT	6
SQXYVT	1

(Many different possibilities)
2 or more correct flows in table

M1

Table correct (adding to 37)

©2025 Exam Papers Practice. All rights reserved.

A1

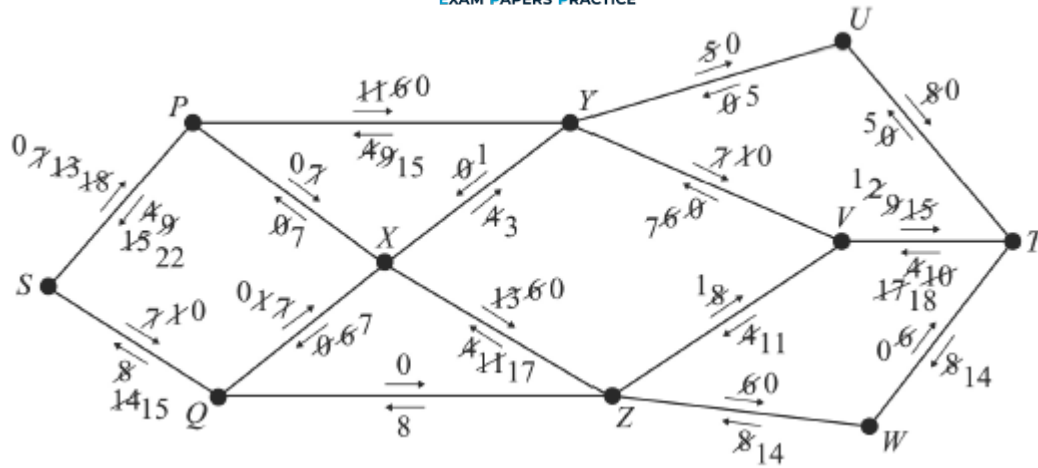
At least 2 flows augmented on diagram

M1

Correct forward and backward final flows

A1

4



Other possibility for ZV, VT, ZW and WT

- (e) Flow from Y to X is 3

B1

1

[13]

Q14.

- (a) (i) $17 - 9 + 16 + 20 = 44$

B1

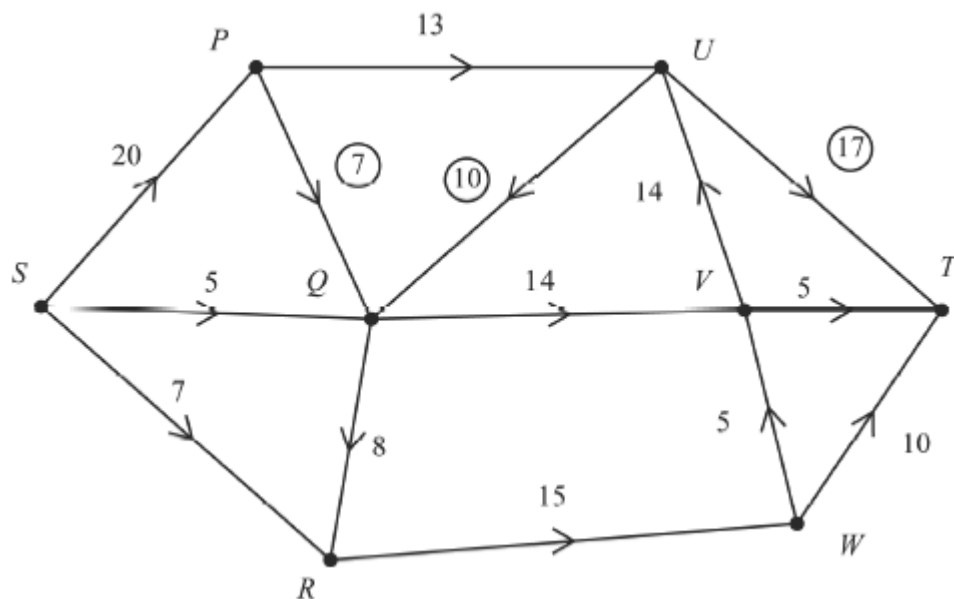
1

- (ii) Max flow ≤ 44

B1ft

1

- (b)



7

B1

10

B1

17

B1

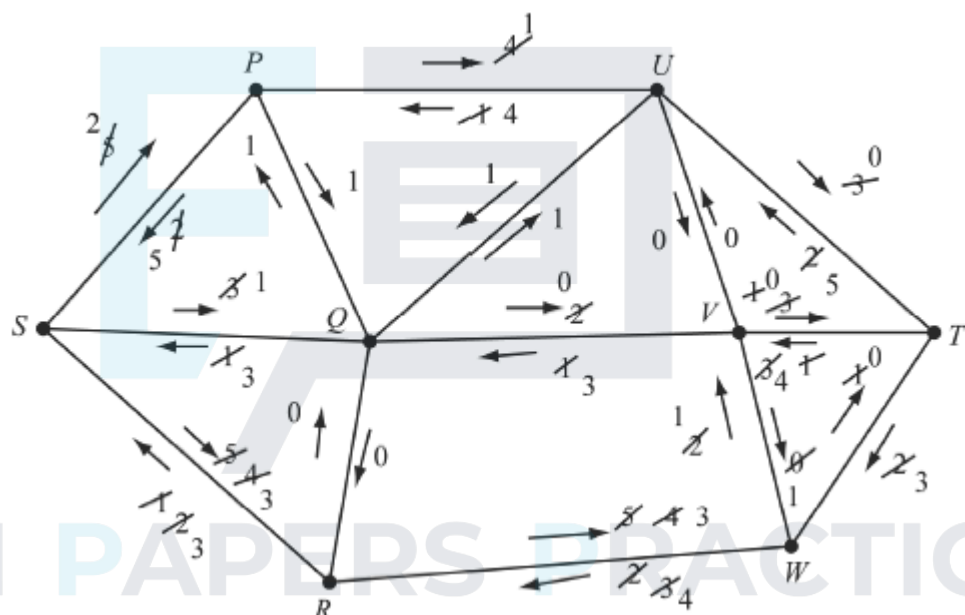
3

- (c) (i) Initial forward and backward flows
5 pairs correct

M1

Correct

A1



©2025 Exam Papers Practice. All rights reserved.

2

(ii)

Path	Additional Flow
SPUT	3
SQVT	2
SRWT	1
SRWVT	1

adjusting flows on network (1 path shown correctly)

M1

correct

A1

additional flow in table

M1

second flow

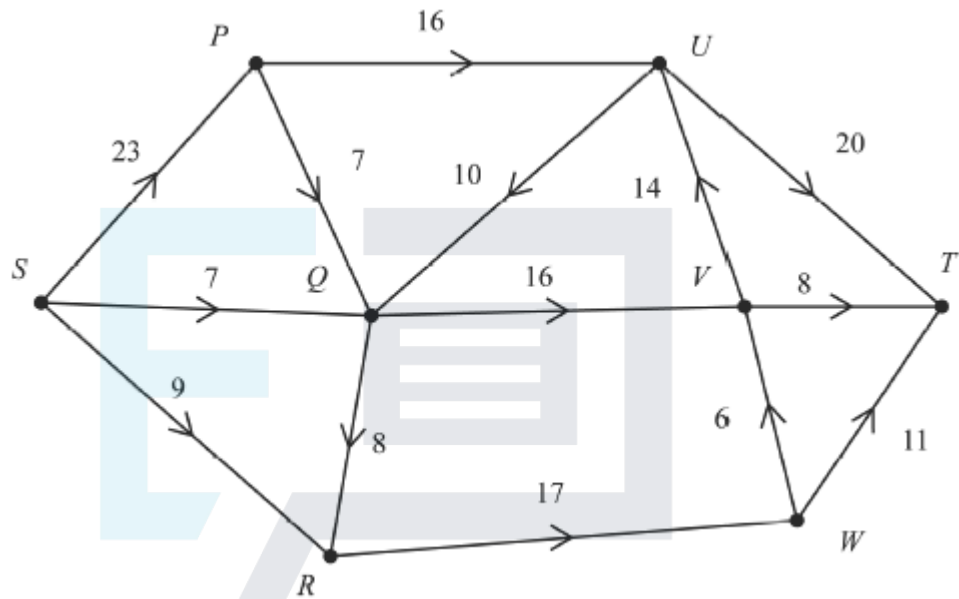
A1

all correct

A1

5

(iii)



Max flow of 39 (several possibilities of final flow diagram)

EXAM PAPERS PRACTICE

B1

1

©2025 Exam Papers Practice. All rights reserved.

[13]

Q15.

(a) (i) $15 + 0 + 14 + 7 + 9 = 45$

B1

1

(ii) Maximum flow ≤ 45
 $\leq$ their value or < 45

M1

correct

A1

2

- (b) *SABT* flow 10
SDET flow 14
SFT flow 9
one correct

B1

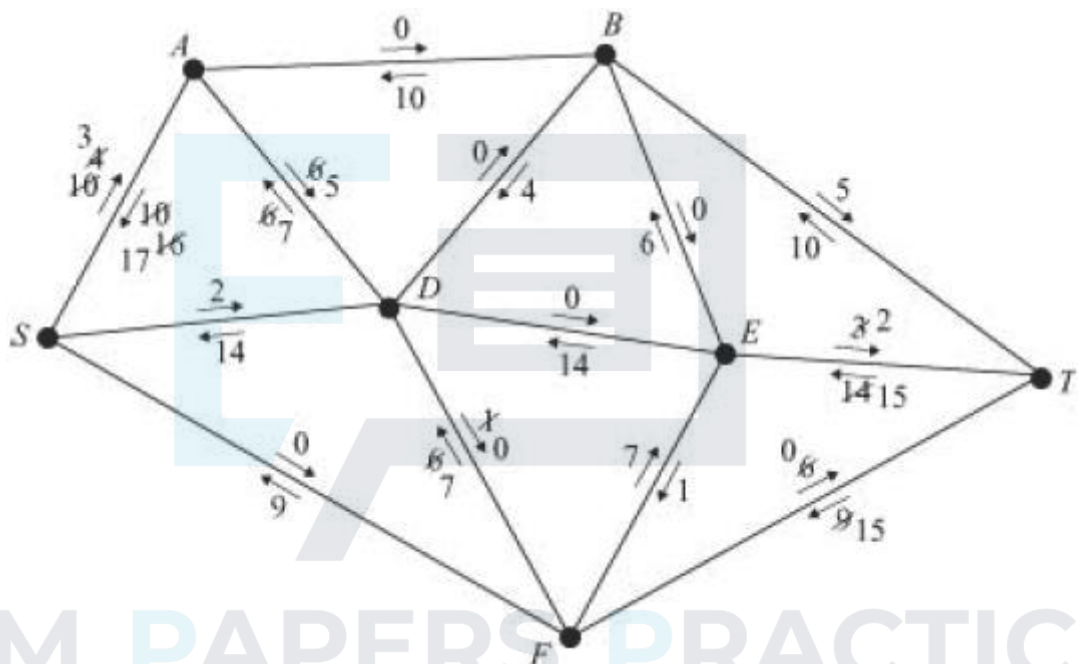
two more correct

B1

(may appear in table below)

2

- (c) (i)



Additional route
 ©2015 Exam Papers Practice. All rights reserved.

M1

with correct flow

A1

one more correct route and flow

A1

table complete

correct total flow of 40

A1

correct use of potential and used flows

on network (may use double edges)

M1

values correctly updated

strict

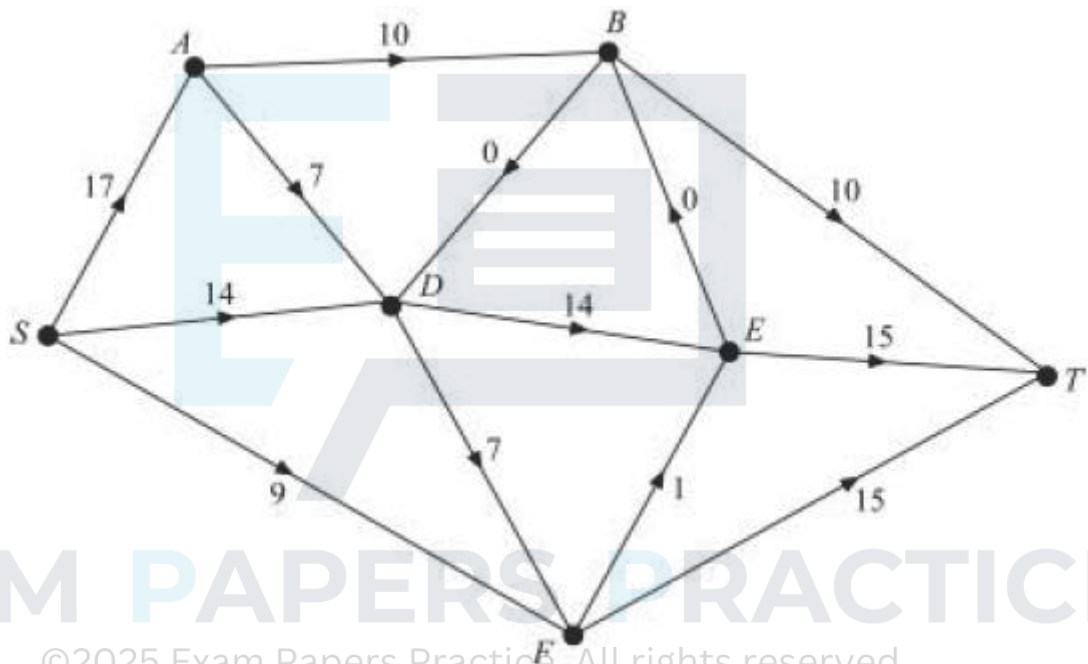
A1

6

Route	Flow
<i>SABT</i>	10
<i>SDET</i>	14
<i>SFT</i>	9
<i>SADFT</i>	6
<i>SADFET</i>	1

several possibilities

(ii)



Maximum flow = 40

B1

Network showing flow of 40

B1

2

(iii) Cut through saturated arcs
AB, BD, DE, DF, SF

M1

Minimum cut shown to be 40 with statement
linking to maximum flow

A1

2

Q16.

(a) (i) $5 + 8 + 16 - 3 = 26$

B1

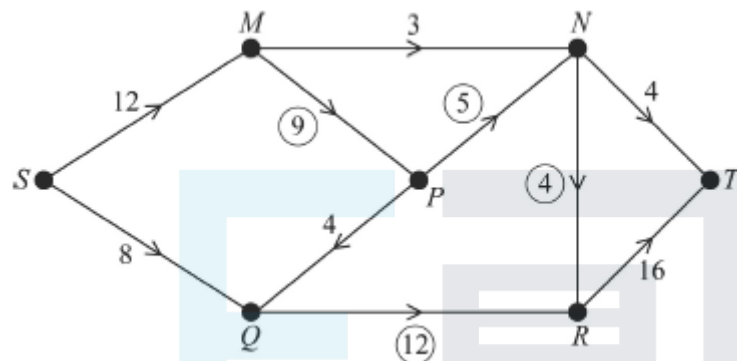
1

(ii) $\text{Max flow} \leq 26$

E1ft

1

(b)



$MP - 9$

B1

$PN - 5$

B1

$NR - 4$

B1

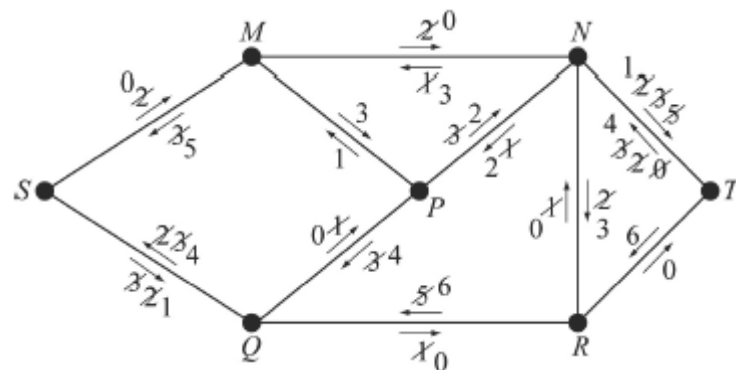
$QR - 12$

©2025 Exam Papers Practice. All rights reserved.

B1

4

(c) (i)



initial flow – forward and backward
6 pairs correct

M1

correct

OM 2 & 3; MN 2 & 1
 NT 5 & 0; MP 3 & 1
 SQ 3 & 2; PQ 3 & 1
 PN 3 & 1; QR 1 & 5
 NR 2 & 1; RT 0 & 6

A1

2

(ii) Adjusting flows on network

M1A1

Path	Flow
SMNT	2
SQPNT	1
SQRNT	1

First correct path and flow

B1

Second correct

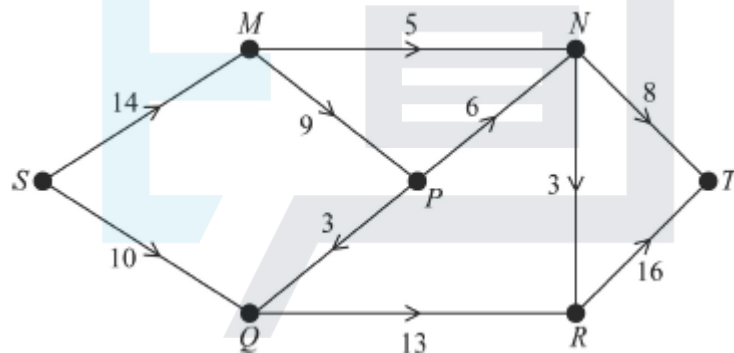
B1

Rest

B1

5

(iii)



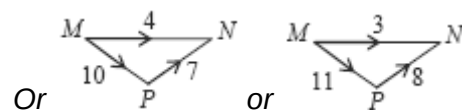
6 flows correct

M1

all correct

©2025 Exam Papers Practice. All rights reserved.

A1



2

[15]

Q17.

(a) D

B1

1

(b) $(17 + 25 + 35 + 13 + 12 + 13 = 115)$

B1

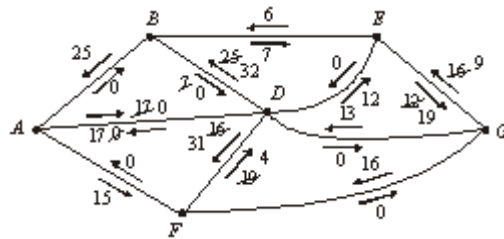
1

(c) $ABD_{\max} = 25$; $GED_{\max} = 12$

B1B1

2

(d) (i)



Forward and backward flows

M1

Adjusting flows on diagram

M1

Routes and flows in chart

M1

One correct other than ABD, GED

A1

Another correct

A1

Route	ABD	GED	GFD	GD	AD	AFD	GED
Flow	25	12	16	13	17	15	7

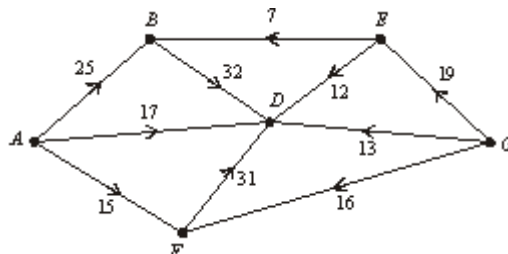
All correct

©2025 Exam Papers Practice. All rights reserved.

6

(ii) Total = 105
Max flow

B1



B1

2

(iii) Cut through AF, AD, BD, DE, DG, and GF

Through 3 saturated arcs (fairly generous)

Correct

M1

A1

2

- (e) Reduce max flow by their *EG*
changing 19 to 15

Reduce by 4 since everywhere else saturated

M1

⇒ New max = 101

Correct answer ⇒ 2 marks

A1

2

[16]

Q18.

- (a) (i) $9 + 7 + 0 + 9 + 13 = 38$

B1

1

- (ii) Maximum flow is less than or equal to 38
< their value of cut

M1

≤ 38
M0 for "equals" their cut

A1ft

2

- (b) *SUYWT* flow of 9

B1

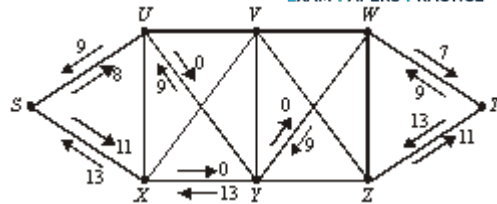
SXYZT flow of 13

B1

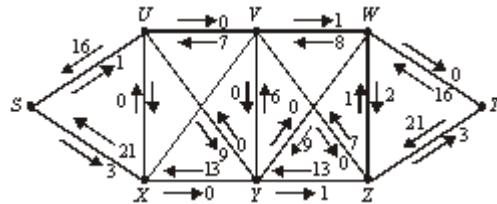
2

- (c) (i) Indicating flows from (b) on network
Preferably as backward flows

M1



with augmentation leading to something such as



Route	Flow
SUYWT	9
SXYZT	13
SUWWT	7
SXVZT	7
SXVWZT	1

Flow augmentation (many possibilities)

SUYWT 7

SXVZT 7

SXVWZT 1

M1A1

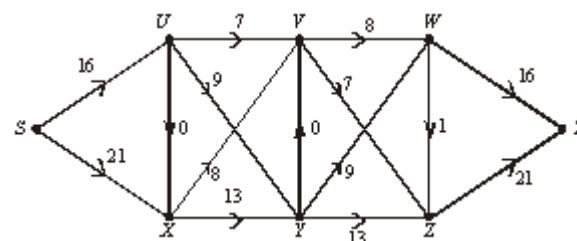
m1A1

A1

6

- (ii) Network showing maximum flow
Several possibilities

B1



Maximum flow is 37

Or $\{S, U, X / V, W, Y, Z, T\}$

B1
2

(iii) Attempt to find cut through saturated arcs

M1

Cut through UV, UY, XV, XY

A1
2

[15]

Q19.

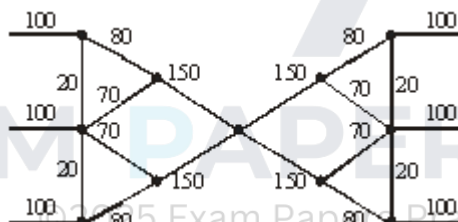
(a) $C_1 = 100 + 80 + 150$
 $= 330$

B1

$C_2 = 80 + 80 + 100 + 80 + 80$
 $= 420$

B1
2

(b)



OE

M1A1A1
3

(c) (i) At I max = 300
 $\therefore$ restriction $\Rightarrow$ max = 120

B1
1

(ii) At H max = 150

M1

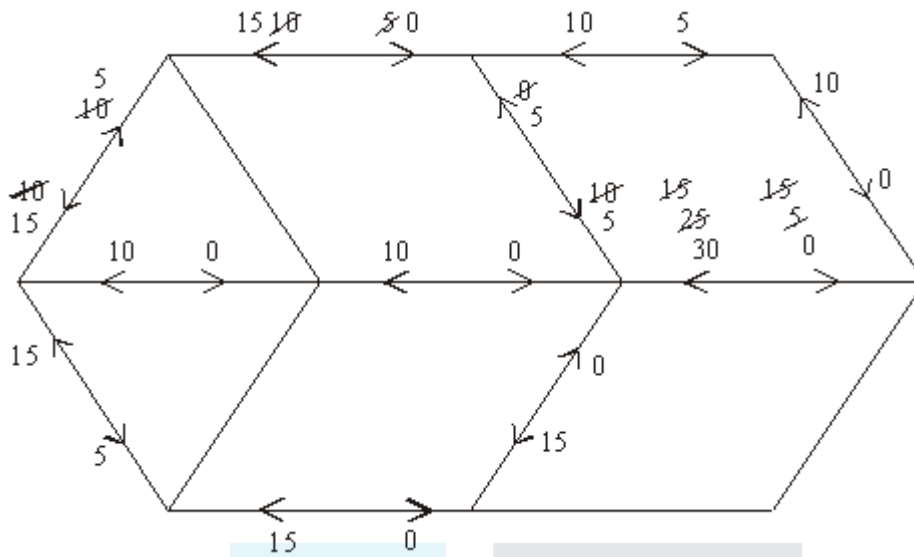
$\therefore$ restriction $\Rightarrow$ max = 270

A1
2

[8]

Q20.

(a)



SCA

first flow

second flow

M1

A1

A1

$\therefore$ maximum flow = 40

B1

4

(b) Cut BE, DG, CF

©2025 Exam Papers Practice. All rights reserved.

B1

1

(c) Out of A – max 50

B1

$\therefore$ from A must be road

Same as J

B1

$\therefore$ join AJ direct

2

[7]

Q21.

(a) $C_1 = 40$

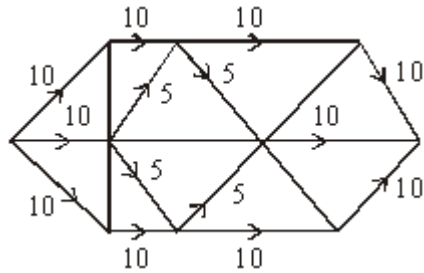
B1

$C_2 = 35$

B1

2

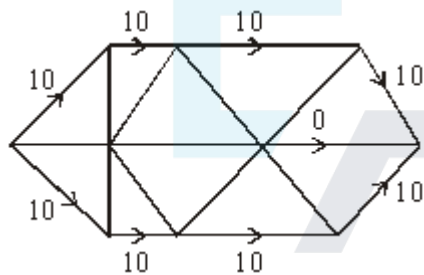
(b)



M1 A1

2

(c)



M1

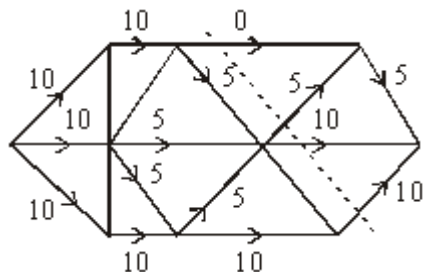
Max flow = 20

©2025 Exam Papers Practice. All rights reserved.

A1

2

(d) (i)



M1

Max flow = 25

A1

2

- (ii) Min cut = 25
Attempted cut

M1

Min cut = Max flow
Correct cut GH , GJ , IJ or $(C_2 - EH)$

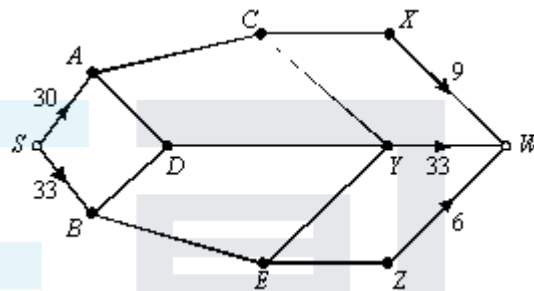
A1

2

[10]

Q22.

(a)



for S and W

2

M1

for five arcs correct ($\geq$)

A1

- (b) $SADYW = 10$

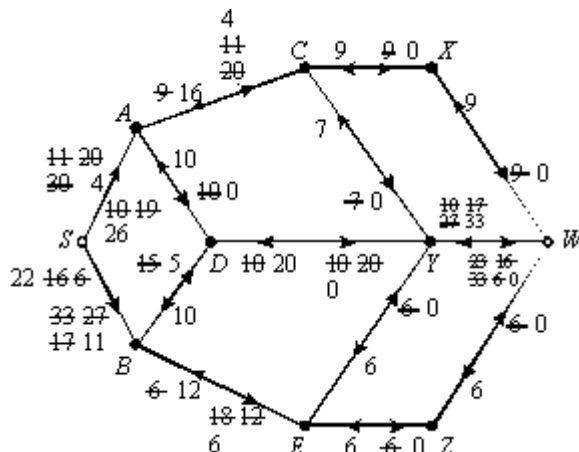
B1

©2025 Exam Papers Practice. All rights reserved.
 $SBEZW = 6$

B1

2

- (c) (i)





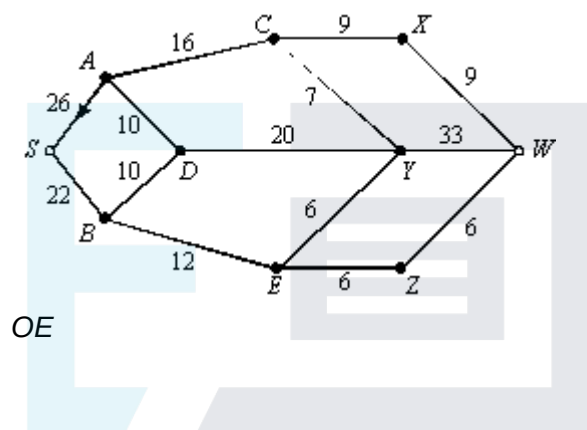
SCA M1

for AC M1

for SA A1

for SB A1

A1



(ii) Maximum flow = 48

Minimum cut CX, YW, EZ
(9) (33) (6)
= 48

Q23.

(a) (i) Flow = 25

(ii) $15 \leq \text{Flow} \leq 25$

(b) At F
Min in = 7

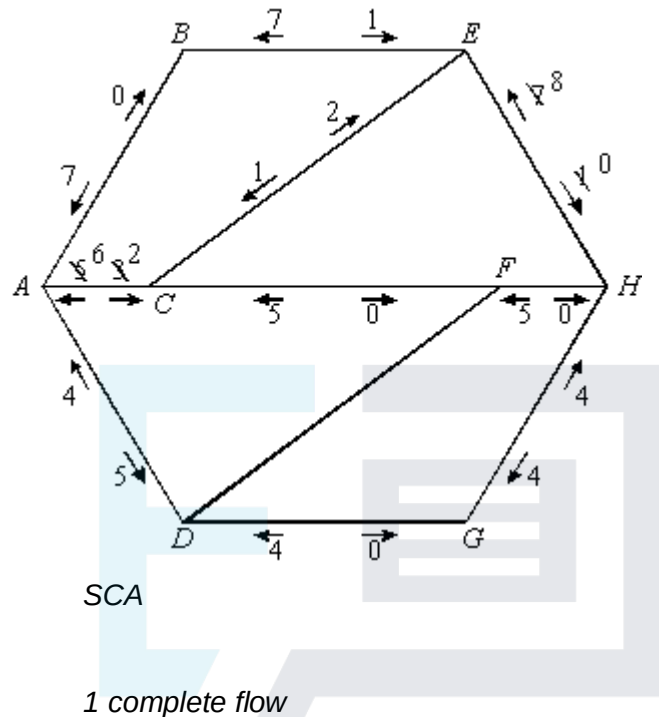
M1

Max out = 5

A1

2

(c) (i)



M1

2nd complete flow

M1 A1

A1

All correct

©2025 Exam Papers Practice. All rights reserved.

A1

5

(ii) Any cut = 17

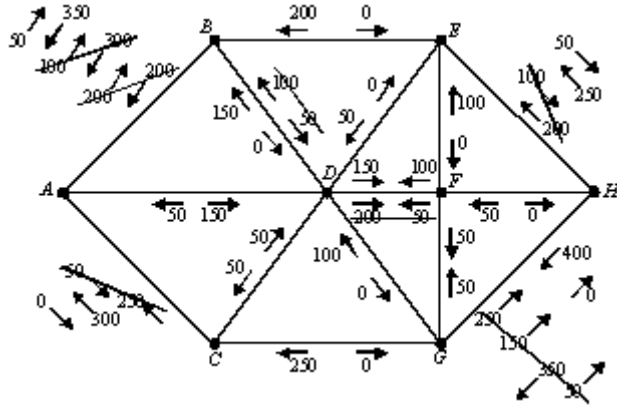
B1

1

[10]

Q24.

(a)



AB or AC started

M1

both initial correct

A1

BE + CG

A1

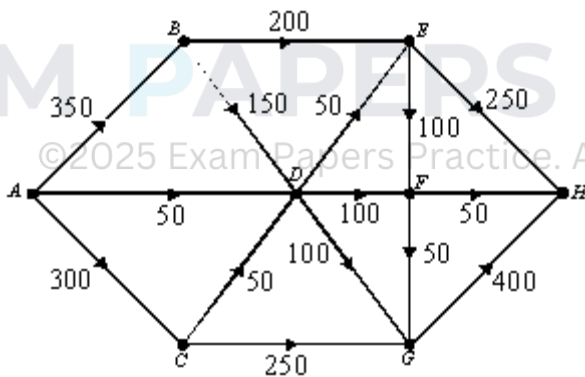
GH, EH

A1

*all correct
o.e.*

A1

Final solution



o.e.

B1

6

- (b) Minimum *BE, DE, FE, FH, GH*
(200) (5) (0) (5) (400)
condone omitting FE

M1

= 700

A1

2

[8]

Q25.

(a) SC, SB, AT

M1 A1

2

(b) (i) SCT 1

M1 A1

2

(ii) $SABCT$ 2

M1 A1

2

(c) any flow $\leq$ any cut
so any flow ≤ 16

B1

1

(d) Choose AT .

B1

The arc must be in the minimum cut in

M1

(a). Also it must be in $\{AT, BT, CT\}$
which is a cut of 17.

A1

3

[10]

Q26.

(a) e.g. $SABT$ 4
 $SACBT$ 2

A1

$SCDT$ 2

A1

$SEDFT$ 3

A1

SEFT 3 (total 14)
 Any two paths
 Remaining paths to
 make total 14
 (or 2 for cao)

A1 5

(b) *EF, ED, CD, BT*
 (3 + 3 + 2 + 6 = 14)

M1 A1 2

(c) (i) Flow in *SE* ≤ flow out at *E* ≤ 3 + 3

B1 1

(ii) Flow into *C* = flow out of *C* ≤ 3 + 2 = 5

M1

So if *AC* has a flow of 4 then *SC* has a flow of at most 1.

A1 2

(iii) If *AC* is saturated the maximum flows in
SA, *SC* and *SE* are 6 (its capacity), 1 (by
 part (ii)) and 6 (by part (i)). So if *AC* is
 saturated the maximum flow out of *S* is
 6 + 6 + 1 = 13 which is less than the
 unrestricted maximum flow.

M1

A1 2

[12]

Q27.

(a) Vertices *S* and *T*

M1

Arcs *SS*₁, *SS*₂, *T*₁*T*, *T*₂*T* and *T*₃*T*
 Capacities 18, 15, 10, 13, 12 (or more) respectively

A1 2

(b) (i) 8 + 2 + 5 + 4 + 12 = 31

B1

1

(ii) $AB \ AC \ DC \ DE$

M1

(or $AB \ CB \ CT_2 \ CE \ DE$)

A1

2

(c) e.g. $SS_1ABT_1T: 8$

M1A1

$SS_1DET_3T: 7$

A1

$SS_2DCET_3T: 5$

A1

$SS_2DACT_2T: 5$

A1

$SS_1ACBT_2T: 2$

$SS_1DCET_2T: 3$

A1

6

(d) All flows $\leq$ all cuts

©2025 Exam Papers Practice. All rights reserved. M1

So, by (b)(ii), all flows ≤ 30 .

Hence the flow of 30 is maximum possible.

A1

2

(e) e.g. For T_1 to get 10 BT_1 will have a flow of 10.

M1

Then, looking at vertex B , max inflow = 10.

Hence BT_2 has 0 flow.

A1

So maximum arriving at T_2 is from CT_2
and ET_2 with a total capacity of 9.

Q28.

- (a) Vertex S , arcs SS_1 and SS_2 with

M1

capacities at least 10 and 11 respectively.

A1

2

- (b) e.g. SS_1ABT 4

M1

SS_1CDT 5

SS_2CDBT 3

A1

SS_2EFT 6

A1

SS_2CEFT 2

A1

A1

5

- (c) S_1A, CD, EF

©2025 Exam Papers Practice. All rights reserved.

M1

Is a cut of 20.

A1

So all flows ≤ 20

A1

3

Q29.

- (a) e.g. SCT 2

M1

$SCBT$ 1

A1

SACBT 2

A1

SABT 2
giving a flow of 7

A1

4

(b) *AB, CB, CT* is a cut of 7

M1

(So the flow of 7 is a maximum)

A1

2

[6]

Q30.

(a) e.g. 300

SABT 100, SADCBT 50

M1

A1 A1

SADCEBT 100, SADCBT 50

A1 A1

Flow of 650

Max of 3 for final flow only

B1

6

(b) *DE, CE, CB, AB*

M1 A1

2

(c) Can augment *SDCT* by 50

B1

New network still has a cut of 700 (*AB, AC, DC, DE*)

M1 A1

3

[11]

Examiner reports

Q1.

The majority of students determined the value of the cut correctly, with the most common errors being numerical slips. Only 51% of students correctly interpreted their value of the cut, with many stating generic comments about the maximum-flow minimum-cut theorem but not making the connection between the value of any cut and the implications for the maximum flow.

Most students spotted the sources and sinks of the network, the most common error being to include a third node which was not a source or sink. However, less than half of the students correctly added arcs to the diagram, with common mistakes being the lack of arrows or correct capacities on each arc.

Q4.

This question proved to be a challenge for students. Students were required to think of the implications of statements and not merely apply an algorithm. Part (a)(i) was answered quite well, but in part (a)(ii), many students failed to state a range for the maximum flow. Some students realised that there had to be an error in part (a)(iii). Fully correct answers to part (b) were rarely seen. Many students highlighted the problem at a vertex other than 'F' and justified their answer by considering maximum flow in and minimum flow out.

Q5.

Most students answered part (a) correctly. Although there were many correct solutions for part (b)(i), a significant number of students failed to give a convincing solution on their diagram. It is essential that students use 'backward' arrows to show existing flows and 'forward' arrows to show potential flows. There was a significant number of students who scored 5 marks in part (b)(i) but failed to score both marks in part (b)(ii). Part (c) was answered poorly. Students were asked to list the edges of their cut; many simply drew a cut on their diagram.

Q6.

©2025 Exam Papers Practice. All rights reserved.

Although many students answered this part correctly, there was a significant number who included values on their edges that corresponded to the initial flow. In part (b), a large number of students started from a 'zero-flow' and then augmented to arrive at the given flow and then stopped. In (b)(i), although there were many correct solutions, a significant number of students failed to give a convincing solution on their diagram. It is essential that students use 'backward' arrows to show existing flows and 'forward' arrows to show potential flows. There were a significant number of students who scored 5 marks in part (b)(i) but failed to score both marks part (ii). Part (c) was poorly answered. Students were asked to list the edges of their cut, many simply drew a cut on their diagram.

Q7.

Part (a) It was surprising to see a large number of able candidates being unable to calculate the correct value of the cut, with many confusing upper and lower capacities in their calculation.

Part (b)(i) Few candidates had trouble in finding the flows along DE and FG .

Part (b)(ii) It was pleasing to see that many more candidates this year were aware of how

to show the potential increases and decreases along each edge. Those who did not do this lost both marks here and two marks in the next part of the question.

Part (b)(iii) It was good to see candidates trying to set out their solution in a logical manner and once again the diagram and table clearly helped. These candidates used the table effectively to show what new flows had been introduced and modified both the forward and backward flows on their network. Marks were awarded for the initial flow and it is very difficult to credit candidates for their original values if they have been obliterated during augmentation; candidates are advised to show the initial flow in ink and then to augment their flow using pencil.

Part (c) Those who stated that the maximum flow was 36 were given some credit for illustrating a flow of 36 on Figure 3. Those with the correct maximum flow were usually able to show a correct flow on the network; those who made a couple of slips were again given some credit for their work.

Part (d) Many candidates were able to find the correct cut with capacity 37, but quite a few who had the correct maximum flow were deceived into using their Figure 3 on which every cut had value 37, rather than looking at the initial network with upper and lower capacities.

Q8.

- (a)
 - (i) Many students were able to show the correct working to give the value of the cut.
 - (ii) Far fewer were successful in finding the three values of the remaining cuts in the table.
 - (iii) Some thought that the “maximum” cut was needed to give the maximum flow and others wrote “maximum flow $\leq$ minimum cut”. The 29 value had been given to help students and this proved valuable for those who had been unable to find the 3 correct values in the table.
 - (iv) Those with a maximum flow of 29 were usually able to draw correct flows along the given edges, although some gave at least one value greater than the capacities on the original network.
- (b) Some missed the vital point of supplementing the flows by 4 along SC and AT and made no further progress. It was good to see many students trying to set out their solution in a logical manner and once again the diagram and table clearly helped. There are still many students unfamiliar with the “labelling procedure” who fail to show both potential forward **and** backward flows on their network. Students are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. Marks are awarded for the initial flow and it is very difficult to credit students for their original values if they have been obliterated during augmentation.

Some credit was given to students who had used flow augmentation correctly but had failed to find a maximum flow equal to 33, if they interpreted their flows correctly on the final diagram.

Q9.

In part (a) most candidates understood how to add a super source and super sink to the network and inserted correct weights on the edges.

In (b) again, few candidates had trouble in finding the maximum flows along the given edges. It was not clear whether some had misread the second route when they gave the flow as 9 instead of 8. For this reason, this error was not penalised heavily in the subsequent work if they eventually found the correct maximum flow through the network.

In part (c) it was good to see candidates trying to set out their solution in a logical manner and once again the diagram and table clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. Marks are awarded for the initial flow and it is very difficult to credit candidates for their original values if they have been obliterated during augmentation.

In (d) those who had a correct maximum flow in part (c) were usually able to find the correct cut with value equal to the maximum flow.

Q10.

Usually after crossing out one or two additional figures, in part (a)(i), most candidates managed to show that the cut had value 97. This(ii) had been intended as a help to find the values of the other cuts in part (a)(ii), but only the best candidates managed to find the 4 correct values in the table. In part (a)(iii), many candidates made statements involving inequalities instead of saying that the maximum flow was equal to the minimum cut. The cut with value 53 had already been provided in the table but several candidates ignored this value.

In part (b)(i), it was good to see candidates trying to set out their solution in a logical manner and once again the insert clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network; however, some candidates started from a flow of zero rather than using their values on Figure 2. Candidates are advised to use the table to show what additional flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. In part (b)(ii), those who had the correct maximum flow and who had used the labelling procedure correctly usually had no trouble in showing this maximum flow on the final diagram.

©2025 Exam Papers Practice. All rights reserved.

Q11.

Part (a) Most candidates seemed unaware of how to calculate the value of the cut correctly because of the upper and lower capacities. Some subtracted 9 instead of 5 from the sum of the other upper capacities; others merely added a set of positive numbers.

Part (b) In contrast, almost everyone scored full marks for finding the value of each of the missing flows along the given edges.

Part (c)(i) Some candidates still seemed unaware of how to represent the potential increases and decreases **from the initial feasible flow**. It requires forward and backward arrows or a duplicate edge; one showing potential forward flow, the other the potential backward flow. The initial flow values are best written in black ink close to the arrows so that any adjustments can be shown in pencil so as not to obliterate the initial flow figures.

Part (c)(ii) A table was provided so that a flow of 2 along *SABT*, for example, could be listed in the table. The potential forward and backward flows along *SA*, *AB* and *BT* could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. If candidates obliterate their values of the potential increases and decreases from part(i) then they risk losing marks for that part. The majority

of candidates found only 2 augmenting paths and were not able to find the correct maximum flow. Those who were not using the labelling procedure correctly sometimes obtained a flow that was impossible; for instance a maximum flow of 29 was seen quite often. It should be noted that candidates are not expected to do flow augmentation with negative flows.

Part (c)(iii) Some credit was given for transferring values from Figure 1 to Figure 2 even if the maximum flow had not been achieved; but full marks could only be scored for a correct flow of 28 through the network.

Part (d) Very few candidates realised the need to consider their saturated arcs in order to obtain the correct cut. Many wrongly considered their flow on Figure 2, where obviously every cut has a value of 28, and consequently wrote down cuts such as the one through *BT*, *DT* and *ET* as their final answer. That is why the question suggested that candidates “may wish to show the cut on the network above” since that network contained all upper and lower capacities.

Q12.

In part (a), most candidates seemed unaware of how to calculate the value of the cut correctly. Many subtracted 17 instead of 10 from the sum of the other upper capacities.

In contrast, in part (b), almost everyone scored full marks for finding the value of each of the missing flows along the given edges.

Some candidates still seemed unaware, in part (c)(i), of how to represent the potential increases and decreases **from the initial feasible flow**. It requires forward and backward arrows or a duplicate edge: one showing potential forward flow, the other the potential backward flow. The initial flow values are best written in black ink close to the arrows so that any adjustments can best be shown in pencil so as not to obliterate the initial flow figures. In part (c)(ii), a table was provided so that a flow of 1 along *SABT*, for example, could be listed in the table. The potential forward and backward flows along *SA*, *AB* and *BT* could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. If candidates obliterate their values of the potential increases and decreases from part (c)(i), then they risk losing marks for that part. The majority of candidates found only 3 augmenting paths and were not able to find the correct maximum flow.

In part (d), those who used flow augmentation correctly usually had no trouble in completing the diagram to show a maximum flow of 44 litres per second.

In part (e), very few candidates realised the need to consider their saturated arcs in order to obtain a minimum cut. Many wrongly considered their flow on Figure 3, where obviously every cut has a value of 44, and consequently wrote down cuts such as *BT*, *ET* and *GT* as their final answer.

Q13.

Part (a) All candidates seemed familiar with the idea of a super-source, *S*, and super-sink, *T*. Although it was possible to assign equal values such as 100 to the capacity on each of the new edges, such a large value often caused problems with flow augmentation. Some incorrectly assigned equal values such as 20 to the new edges. Those using capacities of 22 on *SP* and 15 on *SQ* were usually the most successful.

Part (b) Quite a few candidates made errors in calculating the value of the cut with a common wrong answer being 36. Several incorrectly thought that the maximum flow was

equal to the value of this cut.

Part (c) The values of the initial maximum flows along the given edges were usually correct.

Part (d) It was good to see candidates trying to set out their solution in a logical manner with most candidates showing potential forward and backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values, particularly the initial flow, should be clear to the examiner when such modification is made. It is not wise to totally obliterate previous values when adjustments are made.

Part (e) Some candidates misread this last part and stated the maximum flow through the network. Very few were able to use their final maximum flow diagram to interpret the correct flow from Y to X.

Q14.

In part (a), about half the candidates seemed unaware of how to calculate the value of the cut correctly. In part (b), by contrast, almost everyone scored full marks for finding the value of the missing flows along the given edges.

In part (c)(i), some candidates still seemed unaware of how to represent the potential increases and decreases **from the initial feasible flow**. It required forward and backward arrows or a duplicate edge: one showing potential forward flow, the other the potential backward flow. The initial flow values are best written in black ink close to the arrows, with any adjustments then shown in pencil so as not to obliterate the initial flow figures.

A table was provided in part (c)(ii) so that a flow of 3 along SPUT, for example, could be listed in the table. The potential forward and backward flows along SP, PU and UT could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. There are many correct solutions to this part, and teachers might find it informative to consider different possibilities with their future students in order to illustrate that, whatever solution they obtain, the maximum flow is still the same.

In part (c)(iii), those who used flow augmentation correctly usually had no trouble in completing the diagram to show a maximum flow of 39 litres per second.

Q15.

In part (a), a number of candidates made errors in calculating the value of the cut; a common wrong answer was 39. Several incorrectly thought that the maximum flow was equal to the value of this cut. In part (b), the values of the initial maximum flows were usually correct, and in part (c), it was pleasing to see candidates trying to set out their solution in a logical manner. Some candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made and the final backward flows should be the values transferred onto Figure 1 to give a possible maximum flow. Although quite a few were successful in finding a cut of the same value as the maximum flow, they failed to convince examiners that they had proved that the flow must be a maximum.

Q16.

In part (a), very few candidates seemed prepared for a question with lower capacities as well as upper capacities, and consequently the correct value of the cut was rarely seen.

In contrast, in part (b), almost everyone scored full marks for the missing flows along the given edges.

In part (c)(i), many candidates seemed unaware of how to represent the potential increases and decreases from the initial flow. It requires forward and backward arrows or a duplicate edge: one showing the potential forward flow, the other the potential backward flow. In part (c)(ii), flow augmentation was not widely understood. A table was provided so that a flow of 2 along *SMNT* could be listed in the table. The potential forward and backward flows along *SM*, *MN* and *NT* could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. Candidates who used flow augmentation correctly usually had no trouble in completing the diagram in part (c)(iii) to show a maximum flow of 24.

Q17.

Part (a) was usually correctly answered but a number made errors in calculating the value of the cut in part (b). The values of the initial maximum flows in part (c) were usually correct also. The insert **Figure 4** was intended to help candidates to set out their solution to part (d) in a logical manner. Some candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. It should be clear to the examiner what the previous values were when such modification is made and the final backward flows should be the values transferred onto **Figure 5** to give a possible maximum flow. Very few were successful in finding a cut of the same value as the maximum flow. The final part (e) was well answered by most of those who had found the correct maximum flow.

Q18.

Part (a) A very common wrong answer for the value of the cut in part (i) was 44. This arose from assuming that the edge *VY* contributed 6 rather than zero to the value of the cut. A number of candidates wrote in part (ii) that the maximum flow was equal to the value of their cut and this response scored no marks.

Part (b) It was pleasing to see almost all the candidates finding the correct maximum possible flows along the two routes.

Part (c) The labelling procedure involves showing the flow already in the network, usually as a backward flow, together with the surplus potential flow as two distinct figures on the diagram. Candidates would benefit from seeing how the mark scheme indicates this with a suggested layout. A table was provided to indicate any flow augmentation and it is hoped that candidates will use such a table wisely in any future question of this type. When using the labelling procedure, the numbers on the network should be lightly crossed through so both the original and adjusted values can be clearly seen by the examiner. The setting out of the solution in part (i) needs much greater attention if candidates are to score full marks for using the labelling procedure in future examinations.

The value of the maximum flow in part (ii) was usually stated correctly but not always shown correctly on the diagram in the insert. Although a few made no attempt at the minimum cut, most were able to find this correctly. Some indicated this cut on a diagram; others indicated that the cut passed through the edges *UV*, *UY*, *XV*, *XY*; still others listed the two sets of vertices $\{S, U, x\}$ and $\{V, Y, W, Z, T\}$ separated by the cut; each of these methods scored full marks.

Q19.

This question proved to be a good discriminator with the best candidates scoring maximum marks. Weaker candidates often failed to score the marks in parts (a) and (b). Part (a) showed a lack of understanding of cuts. In part (b) candidates should be encouraged to list the paths that they are augmenting at each stage. Although part (c)(i) was well answered only the best candidates correctly answered part (c)(ii).

Q20.

Good candidates were able to score the majority of marks on this question, with clearly labelled diagrams. Weaker candidates failed to score the marks in part (a). Candidates should be encouraged to list the paths that they are augmenting at each stage. Part (c) was the least well answered question on the paper.

Q21.

This question proved to be far more challenging to candidates than had been expected. Surprisingly, there were very few correct answers to part (a). This showed a lack of understanding as to the value of a cut. Parts (b), (c) and (d) tested candidates understanding of a flow through a network. The numbers were made deliberately easy so as to avoid numerical slips. However, a number of candidates failed to score on these parts. Worryingly, more than one candidate obtained a flow of 40 through the network.

Q22.

Candidates were required to add a super source and a super sink to the given diagram. This was well answered although a number of candidates did not include values on the arcs that at least equalled the total values at points A and B . Part (b) was well answered and to assist candidates they were expected to take these as initial flows in part (c) before using flow augmentation to the diagram. There were many excellent solutions to this question, with working shown on the diagrams, together with a clearly labelled final solution.

Q23.

Candidates' answers given in part (a) were not very accurate, nor were the descriptions given for no possible flow in part (b). In part (b), candidates merely had to refer to the restrictions at F . The final part of the question was well answered but a small number of candidates simply gave a final flow without any indication of method.

Q24.

All the candidates answered this question well using the diagrams provided as working models. There were many different initial flows but eventually the final solutions became clear.

Q27.

Although candidates generally understood what was required, the presentation of the answers was very muddled and hard to follow. In this sort of question, candidates should clearly indicate their final answer, possibly by ringing the flow in each arc or by drawing a separate figure for the final answer.

Q29.

Most candidates successfully produced the flow and cut of 7, but sometimes it was very

hard to follow the working or the final answer in producing the flow. Some candidates followed the good practice of giving their final answer in a diagram separate from the one in which they did their calculations.

Q30.

Most candidates were able to find a maximum flow and minimum cut in parts (a) and (b). They were also able to augment the flow by 50 when an extra arc was added in part (c). However, very surprisingly, few realised that the neatest way of showing that 700 was the new maximum flow was to show that the new network had a cut of 700.



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.